

Indian Market Sales

Exploratory Data Analysis Report

200 sales records | 7 cities | 4 product categories | 2023 - 2025

Data source	MySQL table "data", loaded into pandas
Analysis notebook	Indian_market.ipynb
Tools	Python, pandas, matplotlib, seaborn
Report date	20 September 2026

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1. Executive Summary

This report documents an exploratory analysis of a 200-record sales dataset covering seven Indian cities and four product categories, ordered between 2023 and 2025. The data was pulled from a MySQL table, checked for quality, enriched with date-based fields, and explored through four visual analyses. All figures below are taken directly from the notebook outputs.

200 Sales records	5,181 Total units sold	~565.5M Total sales value (TotalSales)	0 Missing values
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Key findings

- **Hyderabad, Delhi and Chennai lead.** They post the highest summed SalesAmount (3.96M, 3.83M and 3.57M); Mumbai (2.21M) and Kolkata (2.54M) trail. The top region is about 1.8x the bottom one.
- **Chennai and Hyderabad also lead on volume.** Among the five highest-volume regions they account for 22.9% and 22.4% of units, followed by Pune (20.7%), Delhi (18.8%) and Bengaluru (15.3%).
- **Furniture and Clothing drive unit volume.** Furniture sold 1,527 units and Clothing 1,468, together 58% of all units. Grocery is the smallest at 913 units (17.6%).
- **Electronics is the most frequent category** (56 of 200 records) yet sells fewer units per record (22.7) than Furniture (28.3) or Clothing (27.7): many smaller orders rather than large ones.
- **Record counts fall each year:** 85 in 2023, 80 in 2024 and 35 in 2025. The 2025 rows visible in the notebook are all from January-February, so 2025 is likely a partial year rather than a real decline (to be confirmed with the full data).
- **The data is clean.** No nulls in any of the 14 columns, and 18% of records (36) fall on holidays.

Important caveat. In every one of the 10 rows shown in the notebook, $\text{TotalSales} = \text{SalesAmount} \times \text{UnitsSold}$, which indicates SalesAmount is a *per-unit* value. The regional chart sums SalesAmount, so it ranks regions by summed unit values, not by revenue. Revenue by region should be computed from TotalSales (see Section 8).

2. Data Source and Methodology

The dataset lives in a MySQL table named `data`. The notebook connects to the database, reads the full table into a pandas DataFrame with `SELECT * FROM data`, and works with it in memory. Database credentials that appear in the notebook are intentionally omitted from this report.

Workflow followed in the notebook

Step	What was done
1. Load	Imported numpy, pandas, matplotlib and seaborn; connected to MySQL and loaded the table into a DataFrame.
2. Inspect	Checked shape, info, dtypes, head and tail of the data.
3. Summarise	Ran <code>describe()</code> for numeric columns and for categorical (object) columns.
4. Enrich	Converted <code>OrderDate</code> to datetime and added <code>Day_Name</code> , <code>Month_Name</code> , <code>Year</code> and <code>Day</code> columns.
5. Validate	Counted null values in every column.
6. Visualise	Produced four univariate charts: sales by region, top-5 regions by units, units by category, and record counts by category and year.

Dataset structure

The table has 200 rows and 10 original columns; four more were derived in the notebook (marked below).

Column	Type	Description
<code>SaleID</code>	int	Unique record identifier (1-200)
<code>OrderDate</code>	date (text in SQL)	Date the order was placed; converted to datetime
<code>DeliveryDate</code>	date (text in SQL)	Date the order was delivered
<code>ProductCategory</code>	text	Clothing, Electronics, Furniture or Grocery
<code>Region</code>	text	Bengaluru, Chennai, Delhi, Hyderabad, Kolkata, Mumbai or Pune
<code>SalesAmount</code>	int	Sales value per record; appears to be a per-unit value (see caveat)
<code>UnitsSold</code>	int	Units in the order (range 1-49)
<code>IsHoliday</code>	text	Yes / No: whether the order fell on a holiday
<code>PaymentMethod</code>	text	Credit Card, Debit Card, UPI or Cash
<code>TotalSales</code>	float	Total value of the record; equals <code>SalesAmount</code> x <code>UnitsSold</code> in all sampled rows
<code>Day_Name *</code>	text	Weekday name derived from <code>OrderDate</code>
<code>Month_Name *</code>	text	Month name derived from <code>OrderDate</code>
<code>Year *</code>	int	Year derived from <code>OrderDate</code> (2023, 2024, 2025)
<code>Day *</code>	int	Day of month derived from <code>OrderDate</code>

* Derived column created in the notebook. Currency is not stated in the data.

3. Data Quality

Data quality was confirmed with `df.isnull().sum()` after the derived columns were added.

Check	Result
Rows x columns (as loaded)	200 x 10
Missing values	0 in all 14 columns (including the 4 derived ones)
Date parsing	Every OrderDate converted successfully (no NaT), so no derived date field is empty
Distinct values	7 regions, 4 categories, 4 payment methods, 2 holiday flags
Distinct dates	173 unique order dates and 187 unique delivery dates across 200 records
Storage note	Dates arrive as text from SQL and needed explicit conversion to datetime

The dataset is complete and ready for analysis. The most frequent order date, 2024-03-08, appears three times, which does not by itself suggest duplicated records.

4. Descriptive Statistics

Numerical columns

Statistic	SaleID	SalesAmount	UnitsSold	TotalSales
Count	200	200	200	200
Mean	100.5	109,566.40	25.91	2,827,478
Std. deviation	57.88	51,369.85	13.69	2,155,629
Minimum	1	20,769	1	59,353
25th percentile	50.75	67,481.75	16.00	1,019,130
Median	100.5	113,025.00	26.00	2,398,238
75th percentile	150.25	148,610.00	35.25	4,159,576
Maximum	200	199,997	49	9,315,504

- **SalesAmount and UnitsSold are broadly symmetric:** their means (109,566 and 25.9) sit close to their medians (113,025 and 26), so no strong skew.
- **TotalSales is right-skewed:** the mean (2.83M) exceeds the median (2.40M) and the maximum (9.32M) is more than three times the mean. A small number of large orders lift the total.
- **Wide spread:** TotalSales ranges from about 59 thousand to 9.3 million, a natural result of multiplying two variables that each vary widely.

Categorical columns

Column	Unique values	Most frequent	Frequency
OrderDate	173	2024-03-08	3
DeliveryDate	187	2024-10-19	2
ProductCategory	4	Electronics	56 (28%)
Region	7	Hyderabad	36 (18%)
IsHoliday	2	No	164 (82%)
PaymentMethod	4	Credit Card	60 (30%)

Most orders (82%) fall on non-holidays, and Credit Card is the leading payment method at 30% of records. Hyderabad is the most frequent region, appearing in 36 records.

5. Analysis

5.1 Total SalesAmount by region

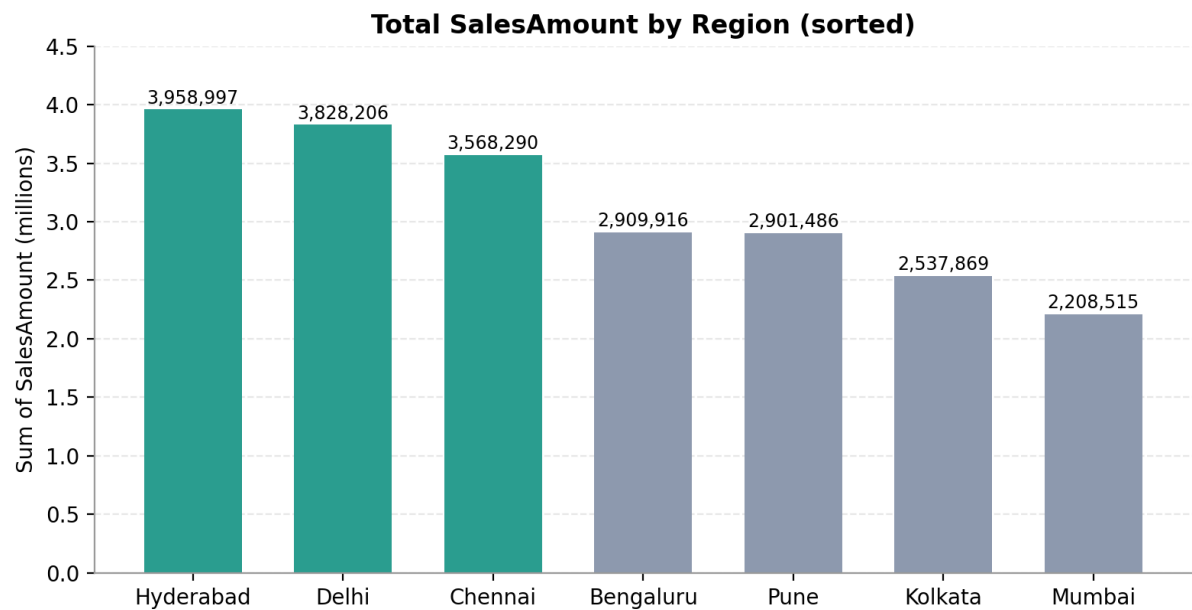


Figure 1. Sum of SalesAmount by region, sorted from highest to lowest (top three highlighted).

Rank	Region	Sum of SalesAmount	Share
1	Hyderabad	3,958,997	18.1%
2	Delhi	3,828,206	17.5%
3	Chennai	3,568,290	16.3%
4	Bengaluru	2,909,916	13.3%
5	Pune	2,901,486	13.2%
6	Kolkata	2,537,869	11.6%
7	Mumbai	2,208,515	10.1%

Hyderabad, Delhi and Chennai together hold about 52% of the summed SalesAmount. Bengaluru and Pune are almost level (a gap of under 10 thousand), while Mumbai records the lowest total. The regional totals add up to 21,913,279, matching the column mean in the summary statistics.

5.2 Top 5 regions by units sold

Share of Units Sold Among the Top 5 Regions

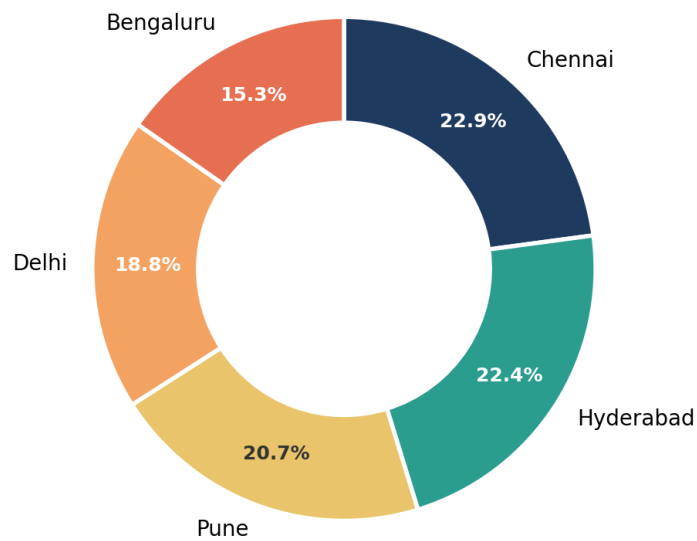


Figure 2. Each region's share of units among the five highest-volume regions (shares sum to 100% of the top five, not of all units).

Chennai (22.9%) and Hyderabad (22.4%) are nearly tied at the top, with Pune (20.7%) close behind. Bengaluru is fifth at 15.3%. Mumbai and Kolkata do not appear, so they sold fewer units than Bengaluru; this is consistent with them ranking last in Figure 1. Note that Delhi ranks second on summed SalesAmount but only fourth on units among these five; this could reflect higher per-unit values or simply more records, and per-region record counts would settle which.

5.3 Units sold by product category

Total Units Sold by Product Category

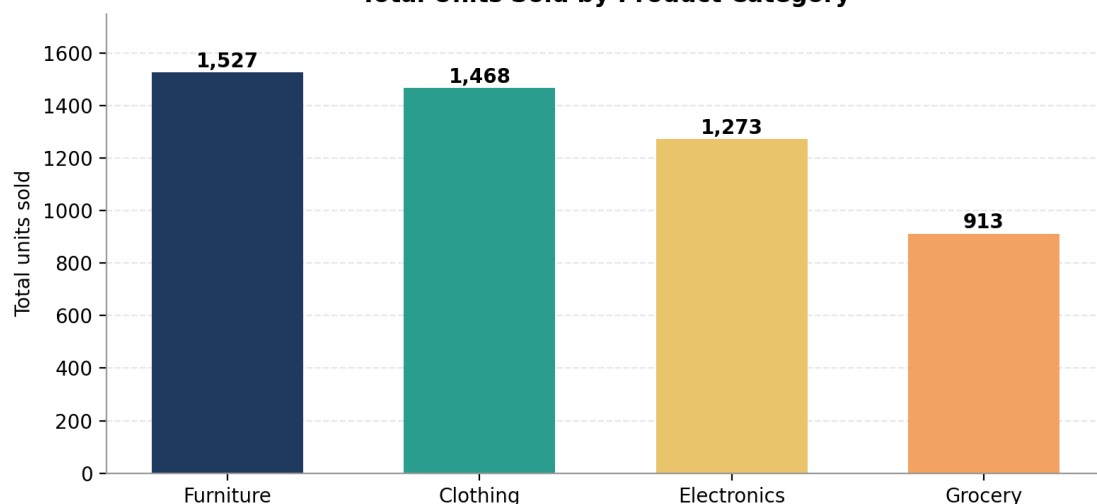


Figure 3. Total units sold per category. The notebook used a connected line; bars are used here because categories have no natural order.

Category	Units sold	Share of units	Records
Furniture	1,527	29.5%	54
Clothing	1,468	28.3%	53
Electronics	1,273	24.6%	56
Grocery	913	17.6%	37
Total	5,181	100%	200

Furniture and Clothing account for 58% of units. Grocery has the fewest records (37) and the fewest units (913).

5.4 Sales records by category and year

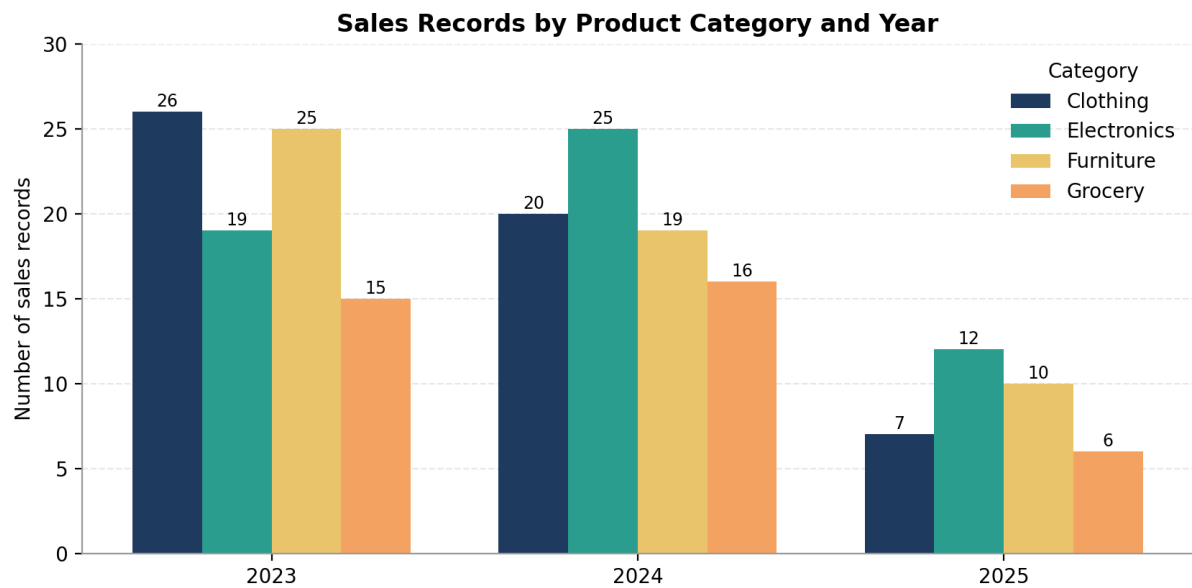


Figure 4. Number of sales records per category in each year. (The notebook titled this "Number of Product Categories"; it counts records.)

Year	Clothing	Electronics	Furniture	Grocery	Total
2023	26	19	25	15	85
2024	20	25	19	16	80
2025	7	12	10	6	35
All	53	56	54	37	200

- **2023:** Clothing (26) and Furniture (25) led; Grocery was smallest (15).
- **2024:** Electronics moved to the top (25) while Clothing (20) and Furniture (19) each dropped.
- **2025:** Electronics (12) remains the largest category, but all counts are low, likely because the year is only partially covered (the 2025 rows shown are all from January-February).
- **Electronics** is the only category whose record count rose from 2023 to 2024 by a large margin (19 to 25); Grocery was roughly flat (15 to 16).

5.5 Derived view: units per sales record

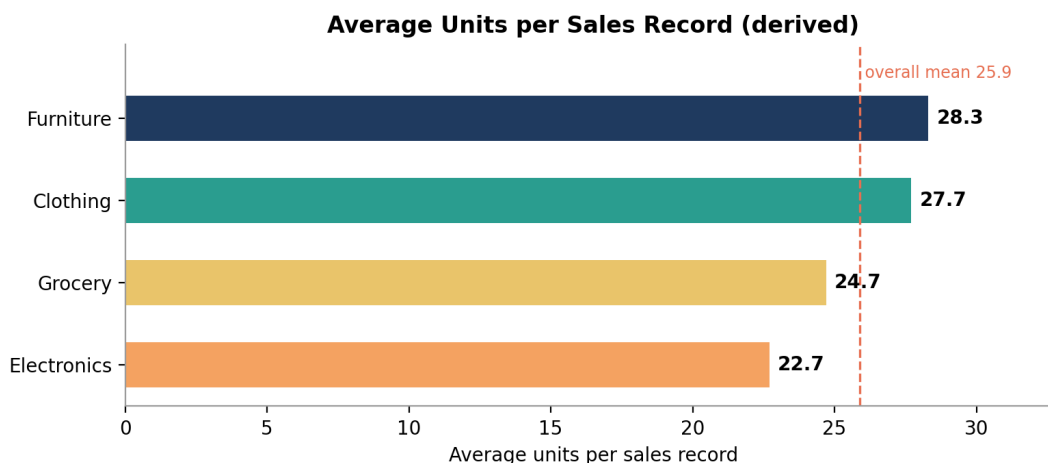


Figure 5. Units sold divided by record count for each category, computed from Figures 3 and 4.

Electronics has the most records but the lowest average order size (22.7 units per record against an overall 25.9). Furniture (28.3) and Clothing (27.7) get more units per record, which is why they lead on total units despite fewer records.

6. Insights and Recommendations

Insight	Suggested action
Hyderabad, Delhi and Chennai lead on summed SalesAmount; Chennai and Hyderabad also lead on units.	Protect stock availability and delivery capacity in these cities; use them as benchmarks for the rest.
Mumbai and Kolkata rank lowest on both sales value and units.	Investigate causes (assortment, pricing, competition, logistics) before setting targets; test promotions there.
Delhi ranks 2nd on SalesAmount but 4th among the top-5 on units.	Compare record counts and average per-unit value by region to see whether pricing, product mix or order volume explains the gap.
Furniture and Clothing sell the most units; Electronics has the most records but smaller orders.	Consider bundles or volume incentives for Electronics to raise order size.
Grocery is the smallest category by records and units.	Assess whether limited range or demand is the cause before investing or de-listing.
Only 18% of records fall on holidays.	Compare holiday vs non-holiday performance before planning holiday-specific campaigns.

These are directional readings from a 200-record sample and univariate views. They generate hypotheses to test rather than firm conclusions.

7. Conclusions

Taken together, the notebook outputs support the following conclusions about the Indian market sales data.

- 1. The dataset is reliable for analysis.** All 200 records are complete, with zero missing values across the 14 columns, and every order date parsed correctly. Nothing needed to be dropped or imputed.
- 2. The business moved a meaningful volume.** The 200 records total 5,181 units and roughly 565.5 million in TotalSales, an average of about 25.9 units and 2.83 million per record.
- 3. Sales value is right-skewed.** The mean TotalSales (2.83M) sits well above the median (2.40M), and the largest record (9.32M) is over three times the mean. A relatively small number of large orders lift the average, so medians give a fairer picture of a typical order.
- 4. Sales are spread across all seven cities, not concentrated in one.** No region holds more than 18.1% of summed SalesAmount, and the top region is only about 1.8 times the bottom one. The three leaders (Hyderabad, Delhi, Chennai) together hold about 52%.
- 5. Hyderabad is the strongest all-round market.** It has the highest summed SalesAmount, the most records (36, or 18%) and the second-largest unit share among the top five (22.4%). Chennai edges it on units (22.9%).
- 6. Mumbai and Kolkata are the weakest markets on both measures.** They rank last on summed SalesAmount and fall outside the top five for units, so they sold fewer units than Bengaluru. These are the clearest candidates for a closer look.
- 7. Delhi shows a mixed profile.** It ranks second on summed SalesAmount but only fourth among the top five on units. Higher per-unit values or a larger number of records could explain this, and the two need to be separated with region-level counts.
- 8. Furniture and Clothing carry the volume.** Together they account for 58% of all units (1,527 and 1,468), while Grocery is the smallest category with 913 units (17.6%) and only 37 records.
- 9. Electronics sells often but in smaller orders.** It is the most frequent category (56 records, 28%) yet averages 22.7 units per record against 25.9 overall, so growth there is more likely to come from larger baskets than from more transactions.
- 10. The category mix shifted from 2023 to 2024.** Electronics rose from 19 to 25 records and became the largest category, while Clothing (26 to 20) and Furniture (25 to 19) declined and Grocery stayed roughly flat. The low 2025 counts most likely reflect a partial year rather than a fall in demand.
- 11. Most sales happen on non-holidays, and no single payment method dominates.** Only 18% of records fall on holidays, and the leading method, Credit Card, accounts for 30% of records. Whether holidays change order size or value has not yet been tested.
- 12. The regional value ranking needs one correction.** Because SalesAmount behaves like a per-unit value, the region chart should be repeated using TotalSales before it is used for revenue decisions. Until then, the regional ranking is best read as directional.

Bottom line. The data is clean and shows a geographically balanced business led by Hyderabad, Delhi and Chennai, with volume driven by Furniture and Clothing and a growing role for Electronics. The main opportunities are lifting order size in Electronics and understanding why Mumbai and Kolkata lag. These findings come from single-variable views of a 200-record sample, so they are best treated as well-supported starting points for deeper analysis.

8. Limitations and Next Steps

Limitations

- **SalesAmount is not revenue.** Sums of SalesAmount (Figure 1) add up per-unit values. Regional revenue should use TotalSales, which the notebook does not chart.
- **Small sample.** With 200 records spread over roughly two years, some categories per year have fewer than ten records, so year-over-year differences are noisy.
- **Probably partial 2025.** If 2025 is only partly covered, year-level comparisons involving it are not like-for-like.
- **Univariate scope.** The four charts each examine one dimension. Relationships (for example region x category) were not analysed.
- **Currency unspecified.** Monetary figures are reported as given, without a currency symbol.

Suggested next analyses

- Revenue (TotalSales) by region, category, year and month.
- Monthly and weekday trends using the Month_Name and Day_Name columns already created.
- Holiday vs non-holiday comparison of units, order value and revenue.
- Payment method mix overall and by region.
- Delivery lead time (DeliveryDate minus OrderDate); the 10 rows displayed in the notebook range from 2 to 10 days.
- Region x category heatmap and a correlation check between SalesAmount, UnitsSold and TotalSales.

The notebook outputs contain summary statistics, ten sample rows and four charts, not the full 200-row table. To extend the analysis with the items above, export the table (for example as CSV) and it can be analysed directly.

Appendix A. Sample Records

First five and last five rows displayed in the notebook (original 10 columns).

ID	Order	Delivery	Category	Region	Sales Amt	Units	Holiday	Payment	TotalSales
1	2024-10-16	2024-10-19	Clothing	Bengaluru	141,958	26	No	Credit Card	3,690,908
2	2023-04-25	2023-05-04	Clothing	Delhi	166,867	32	No	Credit Card	5,339,744
3	2023-01-26	2023-01-28	Clothing	Bengaluru	151,932	6	Yes	Credit Card	911,592
4	2025-01-29	2025-02-03	Electronics	Hyderabad	123,694	32	No	Credit Card	3,958,208
5	2023-10-09	2023-10-18	Furniture	Mumbai	139,879	4	No	UPI	559,516
196	2025-01-09	2025-01-12	Furniture	Delhi	189,695	32	No	Cash	6,070,240
197	2025-01-06	2025-01-14	Clothing	Hyderabad	27,392	23	No	Cash	630,016
198	2023-09-27	2023-10-06	Electronics	Bengaluru	75,680	22	No	Credit Card	1,664,960
199	2024-05-27	2024-06-01	Electronics	Kolkata	197,789	25	No	Credit Card	4,944,725
200	2025-02-19	2025-03-01	Furniture	Pune	70,859	22	No	Debit Card	1,558,898

The rule $\text{TotalSales} = \text{SalesAmount} \times \text{UnitsSold}$ holds for all ten rows (for example, row 1: $141,958 \times 26 = 3,690,908$).

Appendix B. Key Code Used

Derived date columns and regional aggregation, as run in the notebook:

```
df["OrderDate"] = pd.to_datetime(df["OrderDate"], errors="coerce")
df["Day_Name"]   = df["OrderDate"].dt.day_name()
df["Month_Name"] = df["OrderDate"].dt.month_name()
df["Year"]       = df["OrderDate"].dt.year
df["Day"]        = df["OrderDate"].dt.day

sales_by_region = df.groupby("Region")["SalesAmount"].sum()
top_5           = df.groupby("Region")["UnitsSold"].sum().nlargest(5)
units_by_category= df.groupby("ProductCategory")["UnitsSold"].sum()
category_by_year = df.groupby(["Year", "ProductCategory"]).size().unstack(fill_value=0)
```